Indian Statistical Institute, Bangalore

B. Math (hons.) First Year

First Semester - Analysis I

Back Paper Exam | Date: April 12, 2021

Maximum marks: 100 | Duration: 3 hours

1. Prove that $\frac{n!}{n^n}$ converges (*Marks 10*).

2. Determine the convergence of the series $\sum_{n=1}^{\infty}[\frac{1}{n} - \frac{1}{(n+3)}]$ (*Marks 10*).

3. Let $f\colon \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = [x]$, the largest integer not larger than x. Then find $f(x+)$ and $f(x-)$ for any $x \in \mathbb{R}$ (*Marks 20*).

4. Let $f\colon (0,1) \to \mathbb{R}$ be a uniformly continuous function. Prove that $\lim_{n\to\infty} f(\frac{1}{2^n})$ and $\lim_{n\to\infty} f(\frac{1}{3^n})$ exist and are equal (*Marks 20*).

5. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a differentiable function and $g\colon \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = f(x) - f(x+1)$ for all $x \in \mathbb{R}$. If $|f'(x)| \to 0$ as $x \to \pm\infty$, prove that g is bounded on $\mathbb{R}$ (*Marks 20*).

6. Let f be a twice differentiable function on $\mathbb{R}$ with bounded second derivative. Suppose P is a polynomial such that $|P(x)| \leq |f(x)|$ for all $x \in \mathbb{R}$. Find the maximum possible degree of P (*Marks 20*).